4 – 6 March 2013 | **Tutorial Test 3-1** | **2-D Motion /*2-D beweging***

Name/*Naam:* **Student #:**

Monday: 39 a, b

Tuesday: 38 a, b

Wednesday: 35 a, b

Monday: 39 a, b

(a) We note from the graph that the velocity of A is constant at $v_A = 12/6 = 2$ m/s (with two significant figures understood). Therefore, with an initial x_o value of 20 m, car A will be at $x = 28$ m when $t = 4$ s (read off the graph).
This must be the value of x for car B at that time;
Given: $v_{BO} = 12$ m/s
We use $x - x_0 = v_0 t + ½at^2$:

$$28 \text{ m} = (12 \text{ m/s})t + \tfrac{1}{2}a_B t^2 \quad \text{where } t = 4.0 \text{ s}.$$

$$a_B = 2[28 \text{ m} - (12 \text{ m/s})(4.0 \text{ s})]/\, t^2$$

This yields $a_B = -2.5$ m/s^2.

(b) The question is: using the value obtained for a_B in part (a), are there other values of t (besides $t = 4$ s) such that $x_A = x_B$?
Note that the position for A is given as:

$$x_A = x_{0A} + v_A t$$

and the position of B is given by:

$$x_B = x_{0B} + v_B t + ½a_B t^2, \quad \text{where } x_{0B} = 0\,\text{m}$$

Substituting the values for x_{Ao}, v_{Bo} and the acceleration of B from part (a) in the above equations.
The requirement is that the two cars are at the same position (ie $x_A = x_B$)

$$20 + 2t = 12t + \tfrac{1}{2}a_B t^2$$

There are usually two distinct roots for the quadratic equation, unless the discriminant $\sqrt{10^2 - 2(-20)(a_B)}$ is zero. In our case, it <u>is</u> zero – which means there is only one root. The cars are side by side only once at $t = 4$ s.

Tuesday: 38 a, b

We assume the train accelerates from rest ($v_0 = 0$ and $x_0 = 0$) at $a_1 = +1.34\ \mathrm{m/s^2}$ until the midway point and then decelerates at $a_2 = -1.34\ \mathrm{m/s^2}$ until it comes to a stop $\left(v_2 = 0\right)$ at the next station.
The velocity at the midpoint is v_1, which occurs at x_1 = 880/2 = 440m.
(a) $v^2 = v_0^2 + 2ax$ leads to

$$v_1^2 = v_0^2 + 2a_1x_1 \quad \Rightarrow \quad v_1 = \sqrt{2\left(1.34\ \mathrm{m/s^2}\right)\left(440\ \mathrm{m}\right)} = 34.3\ \mathrm{m/s}.$$

The maximum velocity at the midpoint between the stations.

(b) The time t_1 for the accelerating stage is (using Eq. 2-15) $x_1 = v_0t_1 + \frac{1}{2}a_1t_1^2$

$$x_1 = v_0t_1 + \frac{1}{2}a_1t_1^2 \quad \Rightarrow \quad t_1 = \sqrt{\frac{2\left(440\ \mathrm{m}\right)}{1.34\ \mathrm{m/s^2}}} = 25.6\ \mathrm{s}\,.$$

Since the time interval for the decelerating stage turns out to be the same, we double this result and obtain t = 51.2 s for the travel time between stations.

Wednesday: 34 a, b

Let d be the 220 m distance between the cars at $t = 0$,
and v_1 be the 20 km/h = 50/9 m/s = 5.55 m/s speed (corresponding to a passing point of $x_1 = 44.5$ m)
and v_2 be the 40 km/h =100/9 m/s = 11.11 m/s speed (corresponding to a passing point of $x_2 = 77.9$ m) of the red car.
We have two equations based on $x - x_0 = v_0 t + ½at^2$

$$d - x_1 = v_{go}\, t_1 + \frac{1}{2}\, a\, t_1^{\,2} \qquad \text{where } t_1 = x_{r1} / v_{r1}$$

$$d - x_2 = v_{go}\, t_2 + \frac{1}{2}\, a\, t_2^{\,2} \qquad \text{where } t_2 = x_{r2} / v_{r2}$$

With v_{g1o} and v_{g2o} the initial velocity speed of the green car in the two instances,
v_{r1} and v_{r2} is the constant velocity of the red car in the two instances,
$d - x$ is the displacement of the green car when the cars pass each other.

Simultaneously solve these equations and obtain the following results for the: (2 marks for the solving details)

(a) The initial velocity of the green car is $v_o = -8.7$ m/s. The negative indicates that the green car initially moves toward the left i.e. towards the red car.

(b) The corresponding acceleration of the car is $a = -3.3$ m/s^2 (Acceleration towards the left – negative direction).